

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – SHORT ANSWER TEST (30 Questions)

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO4 – Naturalization (BL5)	Assessment:	Assessment Tool 1 – Short Answer Test
Weightage:	40% of CIA	Total Marks:	30 (6 Questions × 5 Marks)
CO4 Statement:	<i>Evaluate visualization designs based on perception, cognition, and effectiveness principles (BL5).</i>		
Student Name:	RENEE VINCENT ASR	Reg. No.:	192424161
Section / Batch:	BTECH AI & DS	Date:	09-082026

Code of Conduct

/ Renee Vincent ASR-192424161 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Renee Vincent

Instructions

- This test contains 6 questions, each carrying 5 marks. Total: 30 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Scatterplots & Correlation Analysis	Correlation, outliers, and relationship identification using scatterplots	1	5
Correlograms & Feature Selection	Correlation matrices and feature importance analysis	2	5
Dimension Reduction (PCA, t-SNE, UMAP)	High-dimensional data visualization and comparison of techniques	3, 4	10
Bubble Charts & Multivariate Visualization	Encoding multiple variables and interpreting limitations	5	5
Hexbin Plots & Dense Data Visualization	Handling overplotting in large datasets	6	5
GRAND TOTAL:			30 Marks

Annexure A – Short Answer Test

1. How does a violin plot provide more information than a box plot when analyzing data distribution?

A violin plot is a statistical visualization used to study the distribution of numerical data across one or more groups. It combines the information of a box plot with a density plot. It shows the median, quartiles, range, spread, and concentration of observations. Therefore, it provides more information about the shape of the distribution than a box plot alone.

a) What does the width of a violin plot represent?

The width represents the probability density or concentration of observations at different values. A wider section means that more observations occur around that value, while a narrow section indicates fewer observations. This allows us to understand not only the center and spread but also where most of the data is concentrated.

b) How can a violin plot help identify multimodal distributions?

A violin plot can clearly show multiple peaks or bulges in a distribution. Two separate wide regions indicate a possible bimodal distribution, while three or more peaks may indicate a multimodal distribution. This can help identify hidden subgroups within the data that may not be obvious from a box plot.

2. How does a pair plot help in exploring relationships among multiple variables in a dataset?

A pair plot, also called a scatterplot matrix, is used to examine relationships between multiple numerical variables at the same time. It displays individual variable distributions along the diagonal and scatter plots between pairs of variables outside the diagonal. It is commonly used in exploratory data analysis.

a) What information is displayed along the diagonal of a pair plot?

The diagonal generally displays the distribution of each individual variable, using a histogram or density plot. It helps identify the variable's central tendency, spread, skewness, and possible outliers or multiple modes.

b) How can pair plots assist in detecting clusters and correlations?

The scatter plots show the relationship between every pair of variables. If points move upward from left to right, there is generally a positive correlation; if they move downward, there is a negative correlation. Closely grouped points can indicate clusters, while isolated points may indicate outliers. Thus, pair plots provide a quick overall view of relationships in multivariate data.

3. What is the purpose of a heatmap in data visualization and analysis?

A heatmap is a graphical technique that represents numerical values through colors. It is especially useful for displaying large matrices of information, such as correlation matrices. Instead of examining many numerical values individually, users can identify important patterns simply by looking at color differences.

a) How are values represented in a heatmap?

Each value is represented by a particular color or color intensity according to a predefined scale. High values, low values, positive correlations, and negative correlations can be represented by different colors. The legend or color scale helps users understand what each color represents.

b) Why are heatmaps useful for identifying patterns and anomalies?

Heatmaps make relationships and variations visually noticeable. Similar colors indicate similar values, while strong changes in color can indicate significant differences. Unusually high or low

values appear as visually distinct cells, helping identify anomalies, extreme values, and unusual patterns. Correlation heatmaps are particularly useful for identifying variables that are strongly related.

4. How does a parallel coordinates plot visualize high-dimensional data?

A parallel coordinates plot is a visualization technique designed for high-dimensional datasets, where each observation contains several variables. It displays all variables simultaneously using parallel vertical axes. Each observation is represented as a connected line passing through the corresponding values on these axes.

a) What does each vertical axis represent?

Each vertical axis represents one variable or dimension of the dataset. For example, a student dataset could contain separate axes for Age, Attendance, Study Hours, Marks, and Grade. Each axis has its own numerical scale, allowing the values of different variables to be displayed together.

b) How can this plot be used to compare observations across multiple variables?

Each observation is shown as a line connecting its values across all variables. By following a line across the axes, we can understand the complete characteristics of an observation. Similar lines can indicate similar observations or groups, while crossing lines can show differences between variables. It is useful for finding patterns, clusters, high-performing observations, and outliers across multiple dimensions.

5. How does a contour plot represent three-dimensional data on a two-dimensional surface?

A contour plot is used to represent three-dimensional data on a two-dimensional surface. Two variables are represented on the X and Y axes, while the third variable is represented through contour lines or different color regions. It is useful when the objective is to study how a value

changes according to two other variables.

a) What do contour lines indicate?

Each contour line connects points that have the same value of the third variable. For example, in a temperature contour map, every point on one line may have the same temperature. Closely spaced contour lines indicate that the value changes rapidly, while widely spaced lines indicate a gradual change.

b) In which applications are contour plots commonly used?

Contour plots are widely used in geography, meteorology, engineering, environmental science, physics, and scientific research. They can represent elevation, temperature, atmospheric pressure, rainfall, pollution concentration, magnetic fields, and other continuous measurements. They help researchers locate regions with high or low values and understand spatial patterns.

6. How does a density plot help in understanding the distribution of continuous data?

A density plot is used to visualize the distribution of a continuous numerical variable. It represents the estimated probability density using a smooth curve. It helps understand the center, spread, shape, skewness, and concentration of the data.

a) How is a density plot different from a histogram?

A histogram divides the data into bins and represents the frequency of observations using bars. A density plot uses a smooth continuous curve to estimate the distribution. Unlike a histogram, the appearance of a density plot is less dependent on bin boundaries, making it useful for comparing multiple distributions on the same graph.

b) What does a peak in a density plot indicate about the data?

A peak indicates a range where the concentration of observations is high. The highest peak generally represents the most common range of values. Multiple peaks can indicate the presence of different groups or populations in the dataset. The position, height, and shape of peaks can also help identify skewness, clustering, and differences between distributions.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)	Criteria	
Conceptual Understanding	30%	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding	Conceptual Understanding	30%
Accuracy of Answer	25%	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer	Accuracy of Answer	25%
Application of Knowledge	20%	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application	Application of Knowledge	20%
Completeness of Response	15%	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response	Completeness of Response	15%
Clarity & Presentation	10%	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand	Clarity & Presentation	

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____